

GMDM Vendor Forum IOP II Population and Serialization Guidelines

Version 1.

October 31, 2025

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GMDM information model

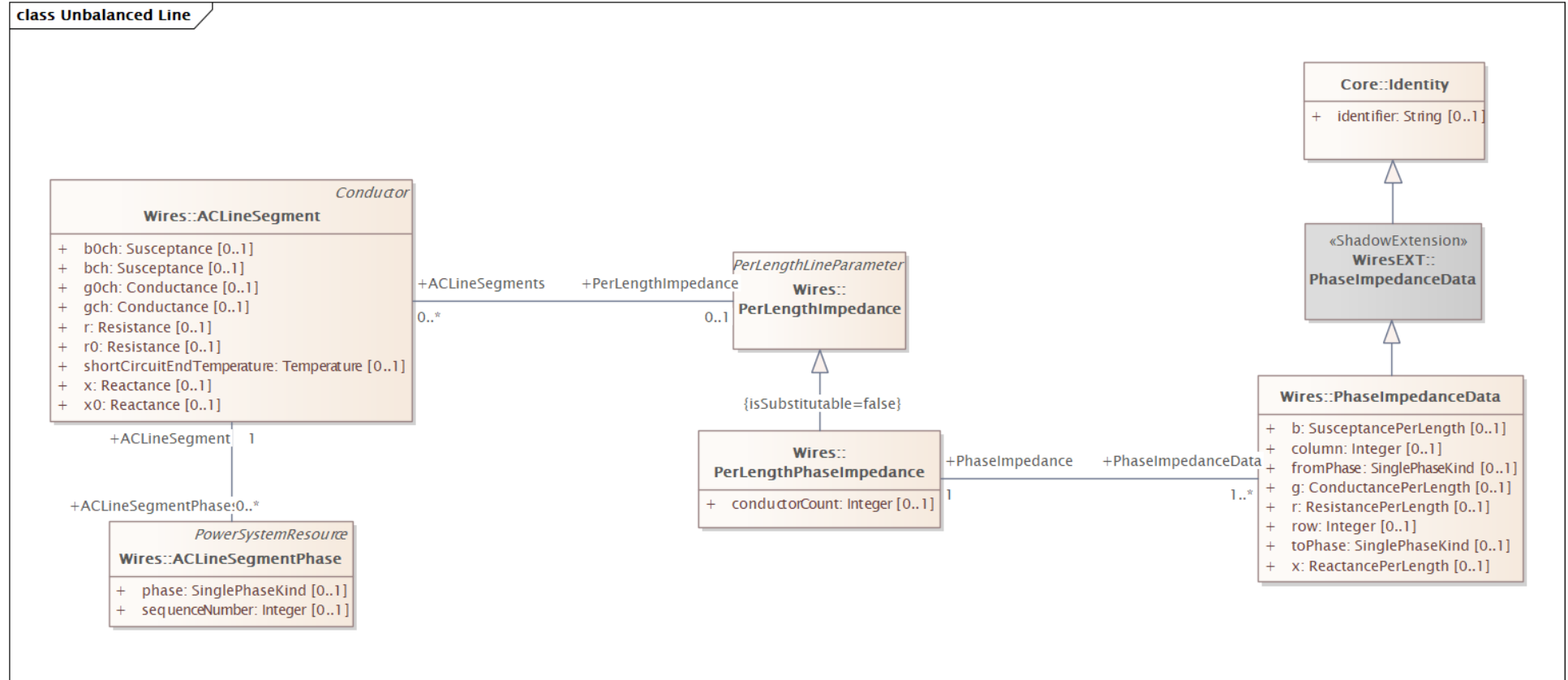
Base

The base model for the GMDM IOP is CIM_Grid18v15_Enterprise14v04_Market04v18.eap. It was the most current stable merged version at the time GMDM profiling work began.

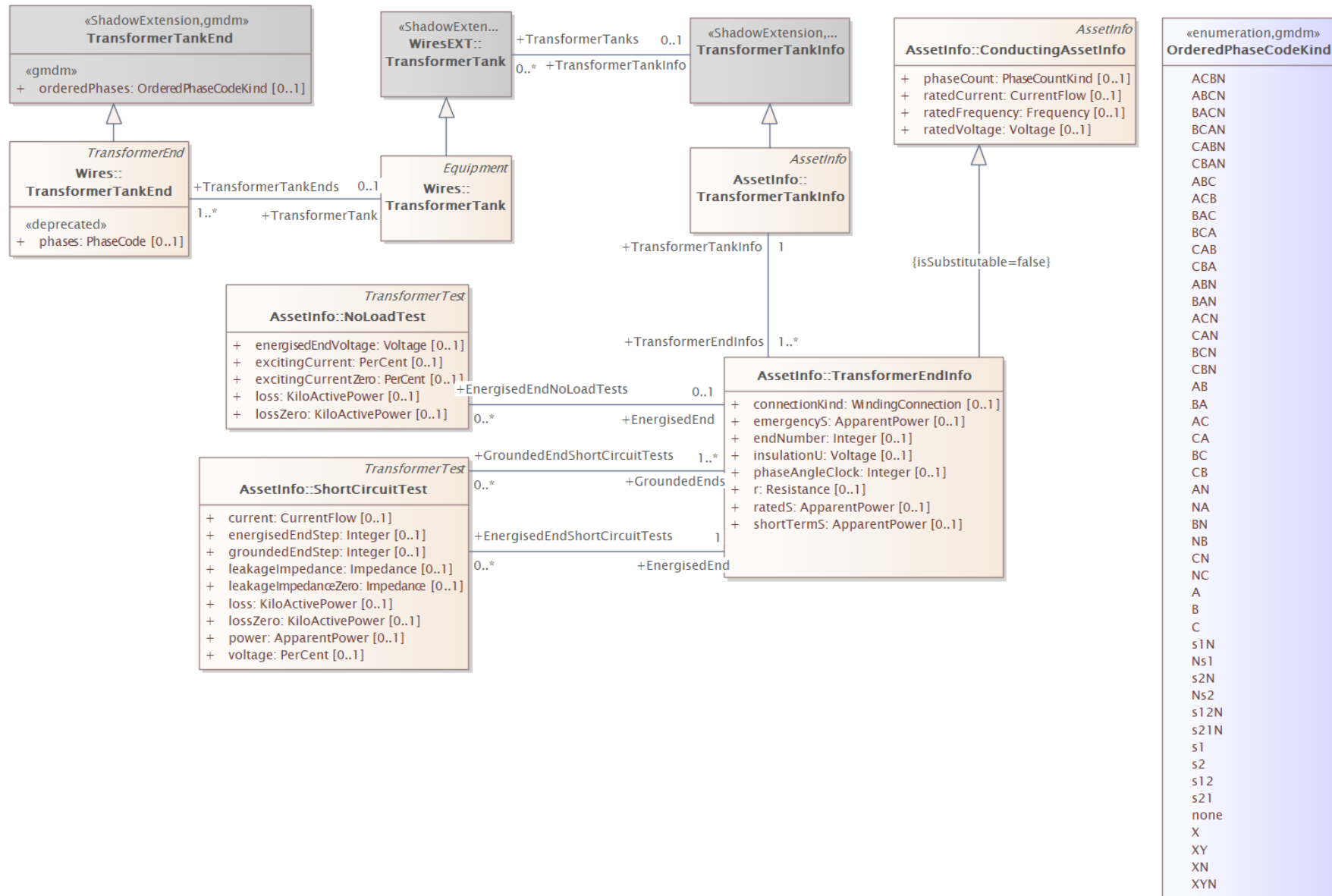
Extensions

GMDM IOP extensions were created using the ShadowExtension approach.

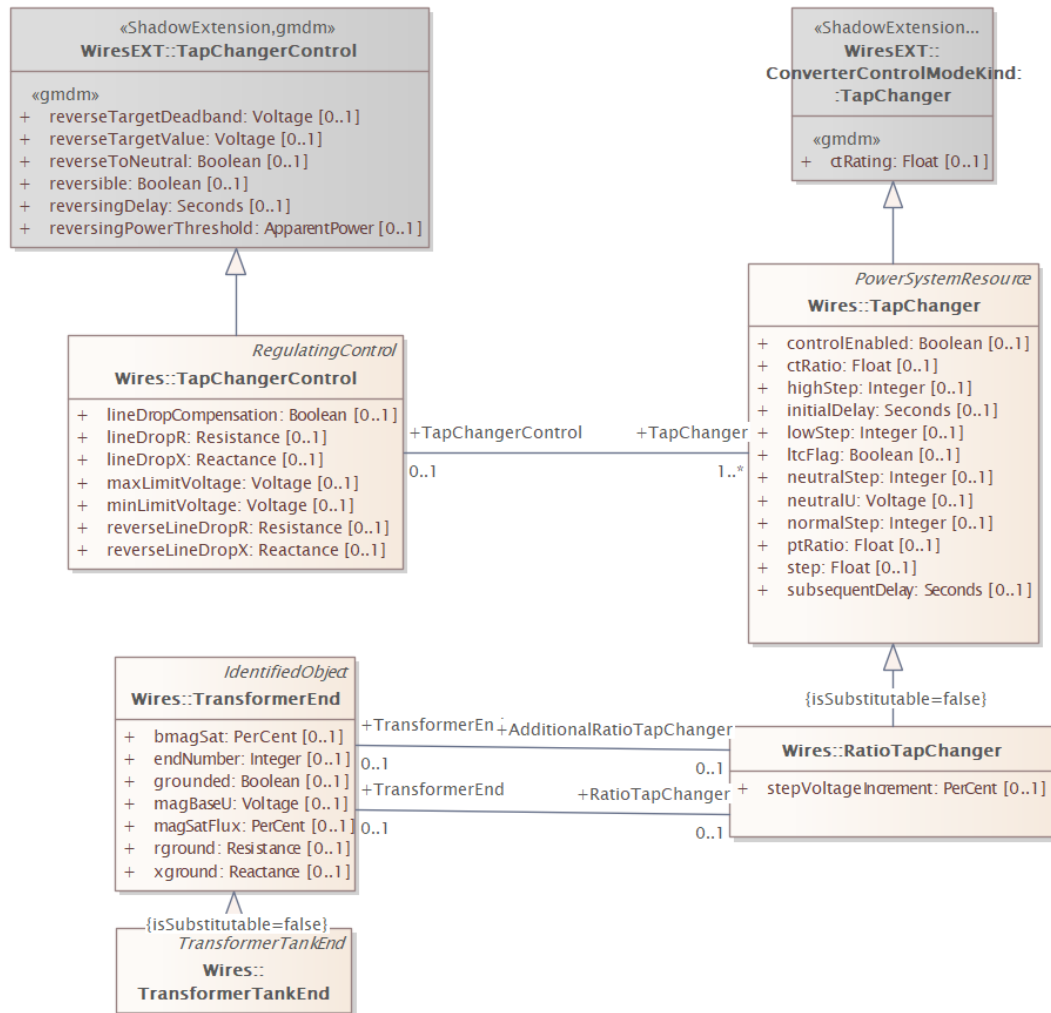
The following diagrams illustrate the GMDM IOP extensions in context with their related classes. ShadowExtension classes are shown in grey.



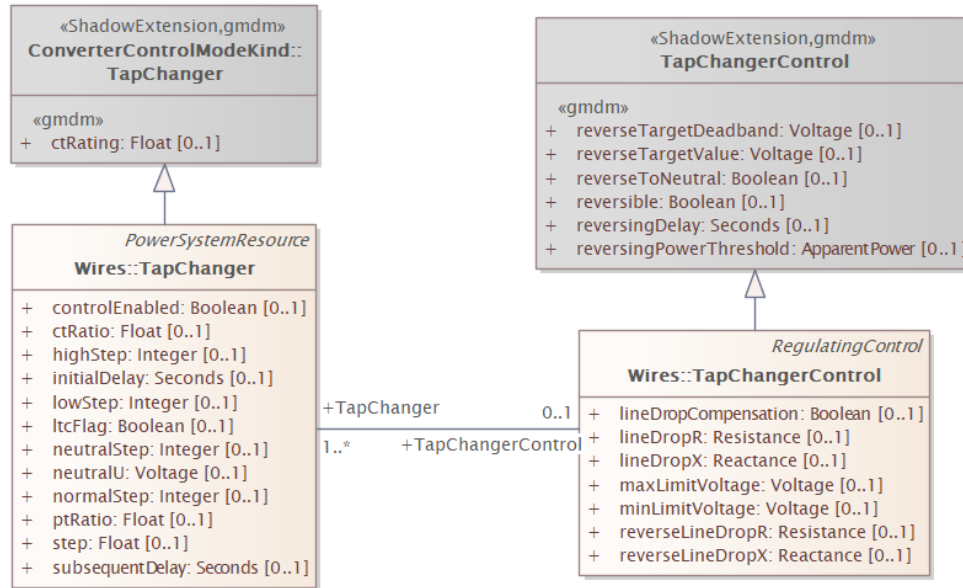
class Transformer Template



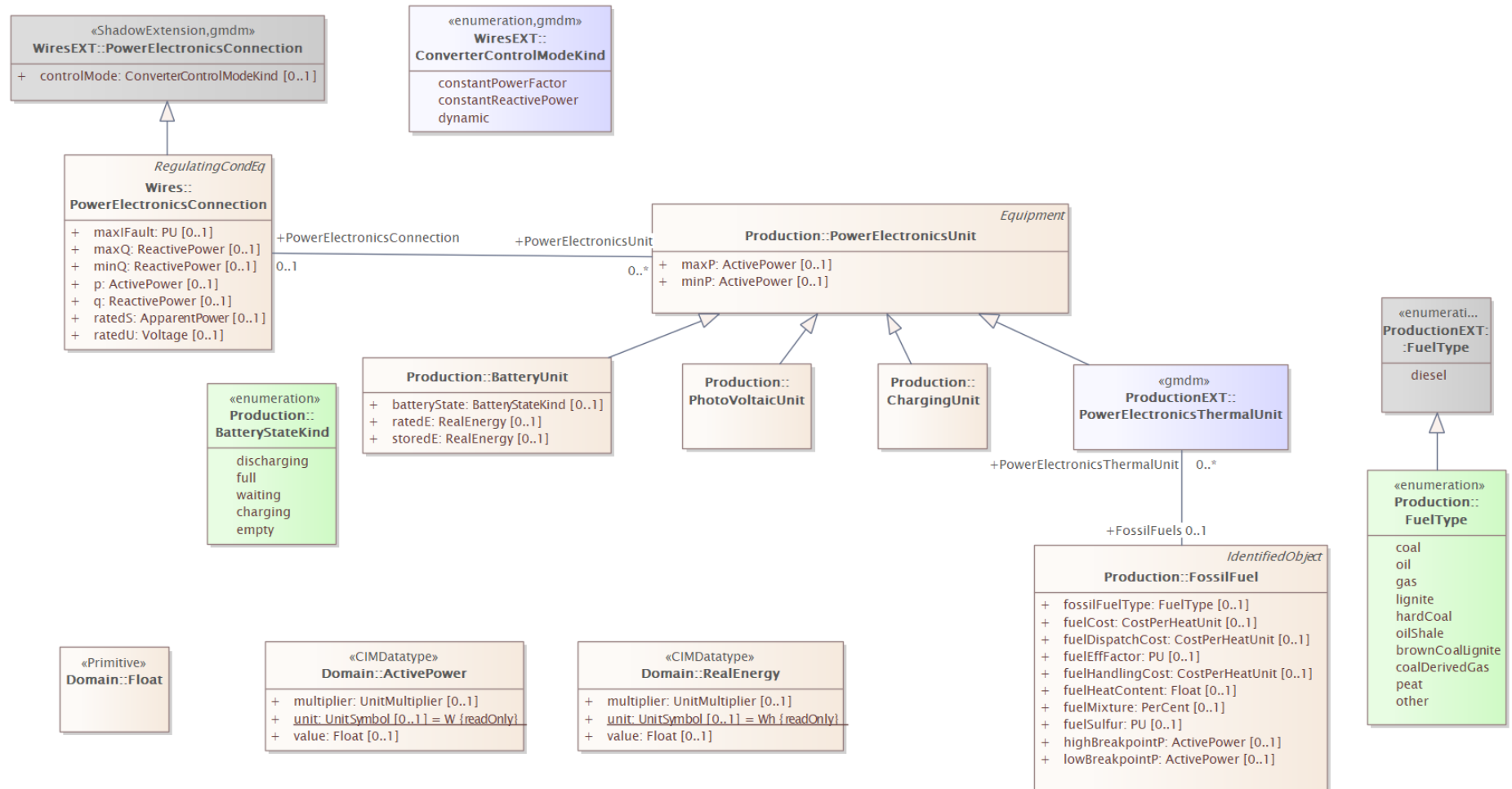
```
class VoltageRegulator
```



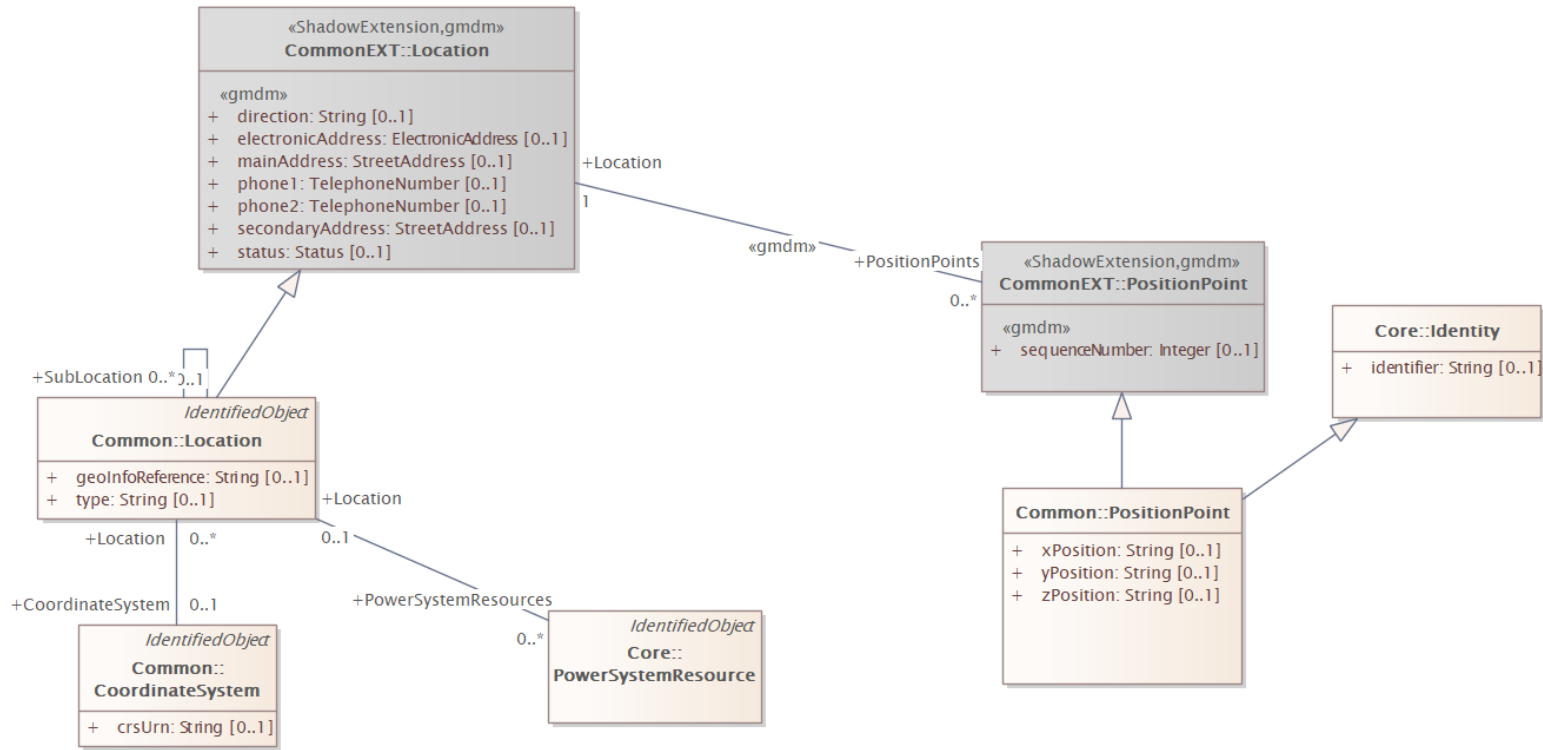
class Tap Changer



class Inverter

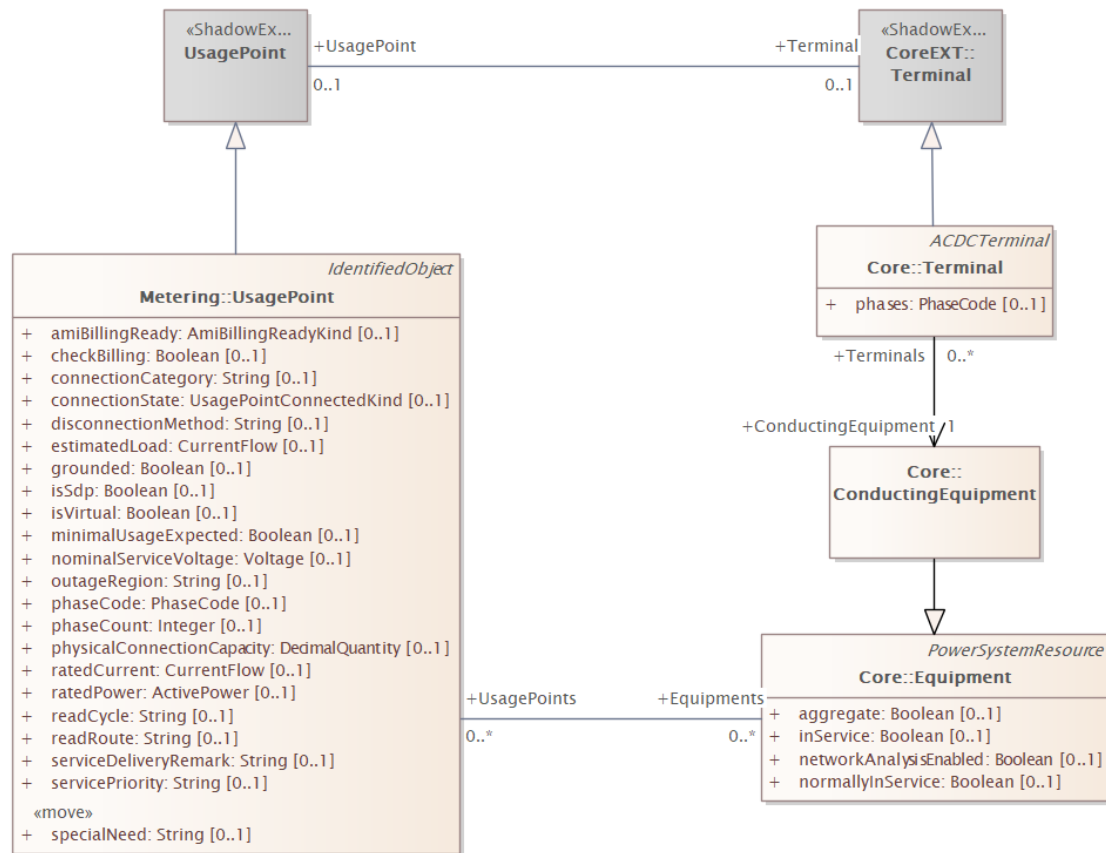


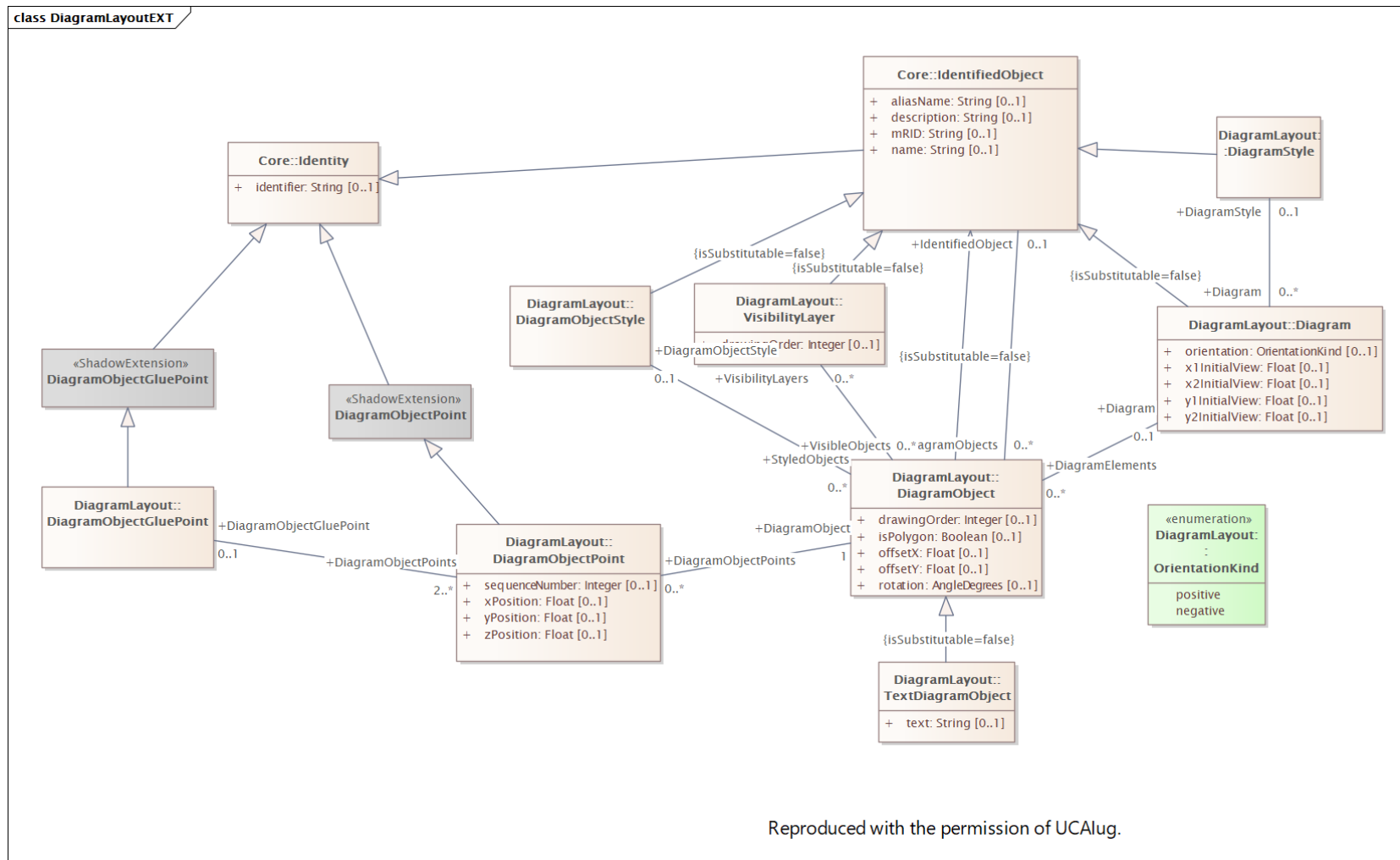
class Location



Will inherit from Identity in next release

class MeteringEXT





- GeographicalLocation (GL) – the geo-spatial location of equipment
- Asset (Asset) – physical equipment characteristics relevant to network analysis or operations (only transformer asset info is part of the GMDM IOP)
- DiagramLayout (DL) – the layout of grid objects on a display (used in the GMDM IOP only for Facilities data)

CIM Datatypes

Typically CIM Datatypes have both their .unit and .multiplier attribute values specified in profiles. For the GMDM IOP, .units and .multipliers are included in the profiles only where values are specified in the underlying canonical UML. The canonical UML has .unit values for all CIM Datatypes, but multiple CIM Datatypes do not have .multiplier values. For CIM Datatypes lacking a .multiplier, the value of the multiplier is assumed to be none.

Population conventions

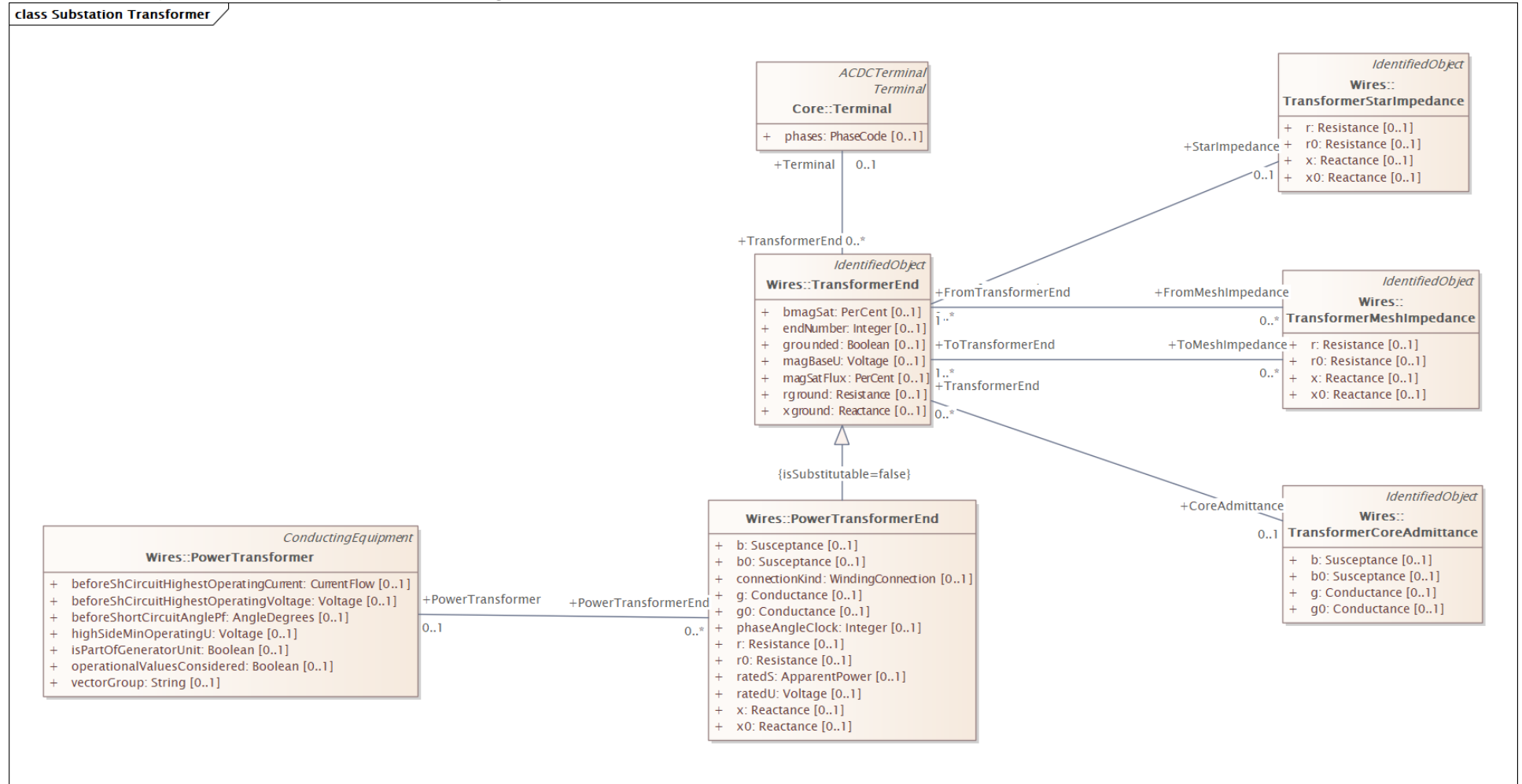
Transformers

There are several types of transformers modelled in the GMDM IOP instance models:

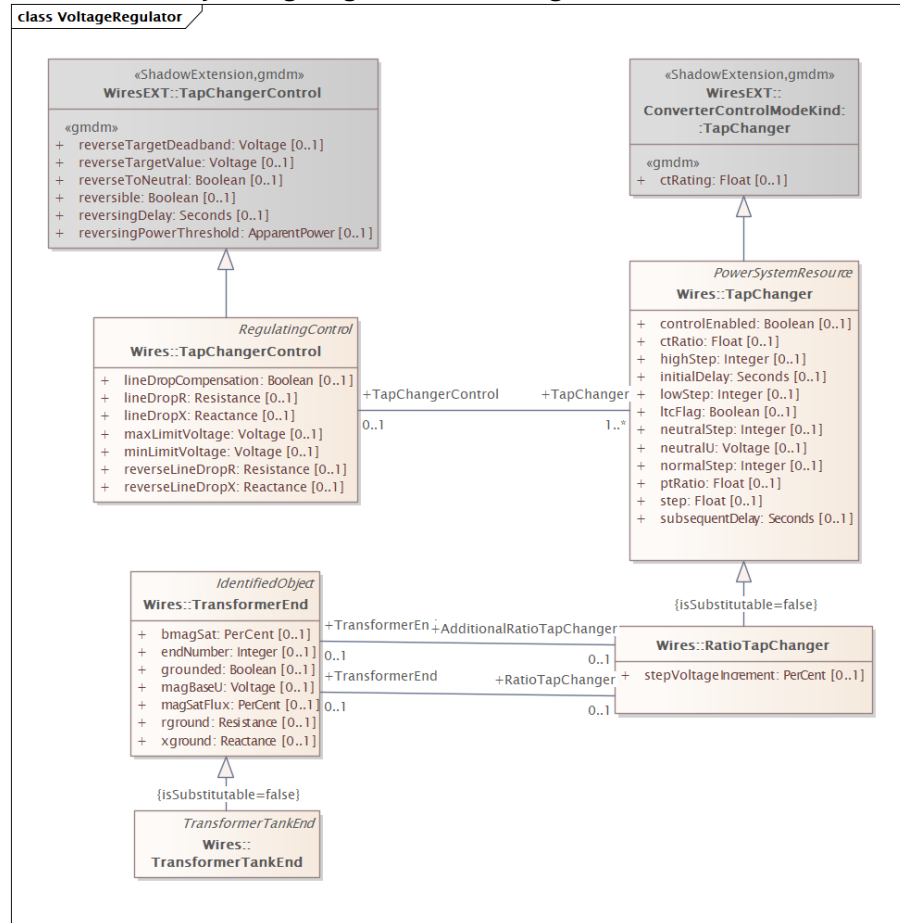
- In SubTransmission Models, there are
 - Substation transformers (115kv to 69kV and 69kV to 12kV), which have no tanks, no tap changers, and no Asset profile information
 - Voltage regulators below the 69kV to 12kV substation transformers, which have 3 tanks, tap changers, and test results supplied in the Asset profile
- In Distribution Models, there are
 - Generator step-up transformers (12kV to a higher voltage), which have no tanks, no tap changers, and no Asset profile information
 - Voltage regulators on feeders (12kV to 12kV), which have 3 tanks, tap changers, and test results supplied in the Asset profile
 - Voltage regulators ??
 - Service transformers (12kV to split 240kV), which have 1 tank,

[illegible]

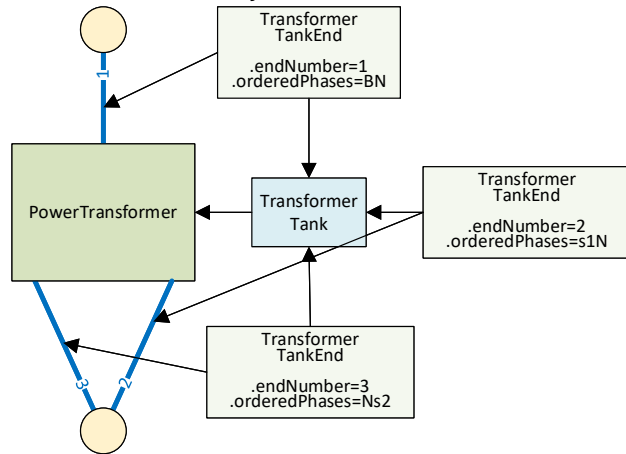
Classes used by substation transformer modelling:



Classes used by voltage regulator modelling:



An illustration of the CN objects used to model a service transformer:



In general, PowerTransformers

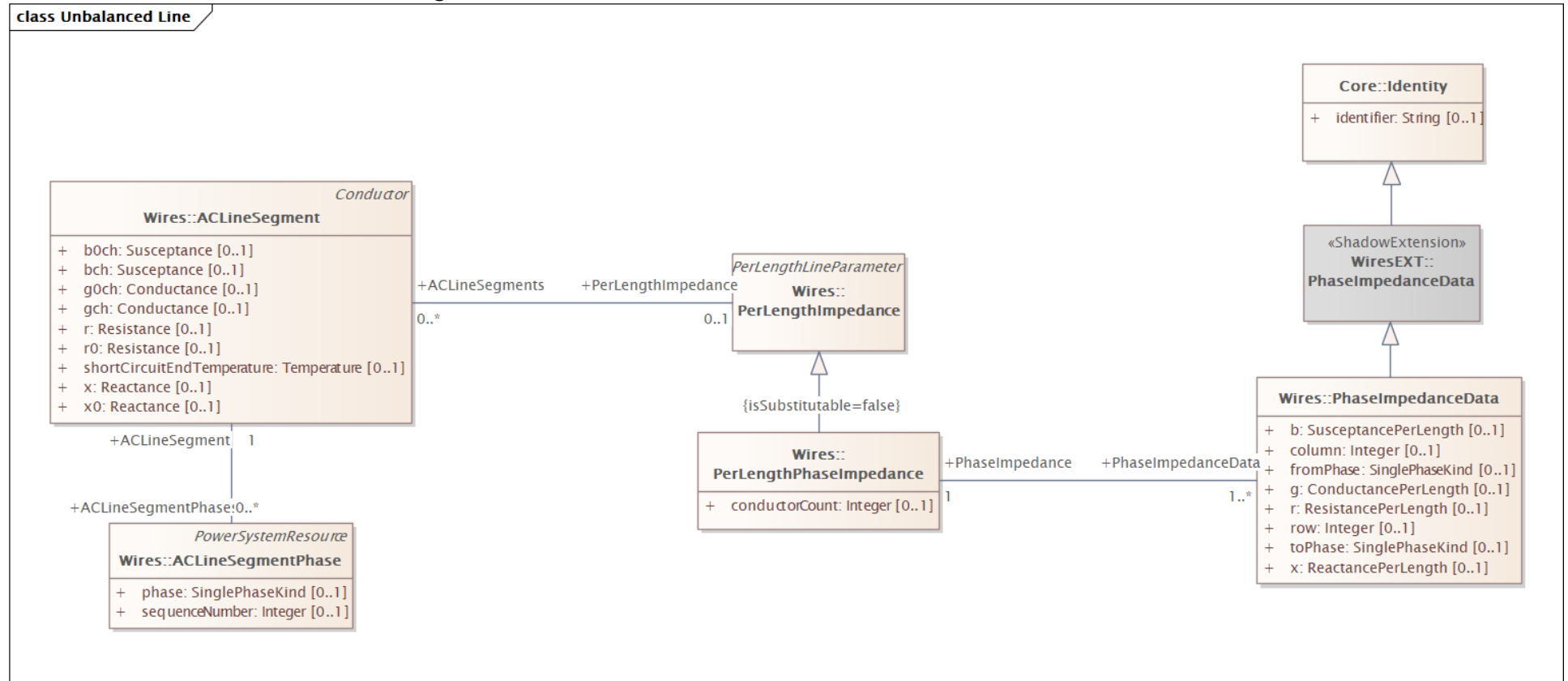
- have TransformerEnds (or TransformerTankEnds) with associations to different VoltageLevels
- are contained in Substations

However, PowerTransformers that are voltage regulators in substations

- have TransformerEnds (or TransformerTankEnds) with associations to the same VoltageLevels
- are contained in VoltageLevels

Unbalanced Lines

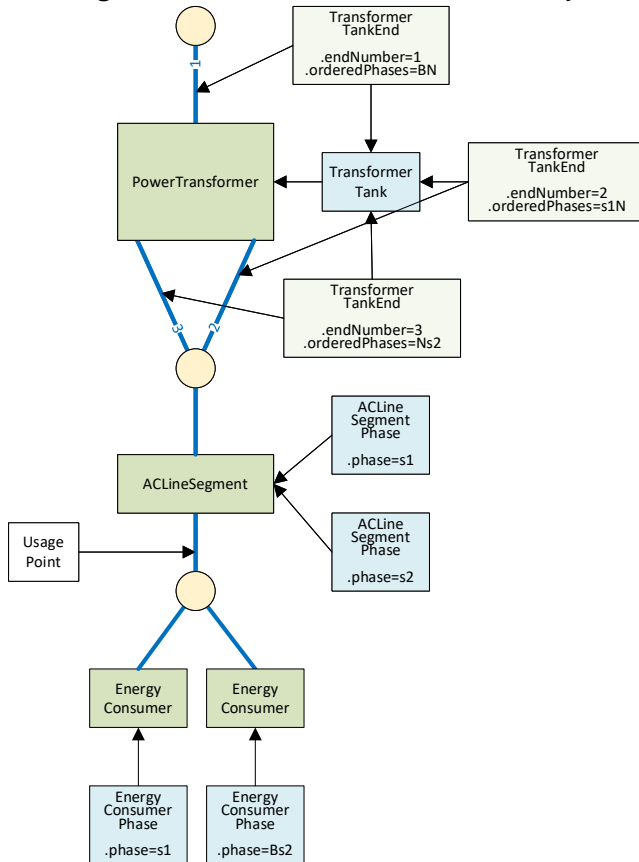
Classes used in unbalanced line modelling:



EnergyConsumers and PhotoVoltaicUnits

Because residential loads are fed with split 240V service an EnergyConsumer is modelled for each split of the phase.

The diagram below illustrates some of the objects used to represent a residential customer supplied from a service transformer:



Containment

A Conducting Equipment may have both an association to BaseVoltage and an association to VoltageLevel. GeographicalRegion and SubGeographicalRegion are not in the GMDM Connectivity profile. ConnectivityNodes must be contained in a ConnectivityNodeContainer subtype.

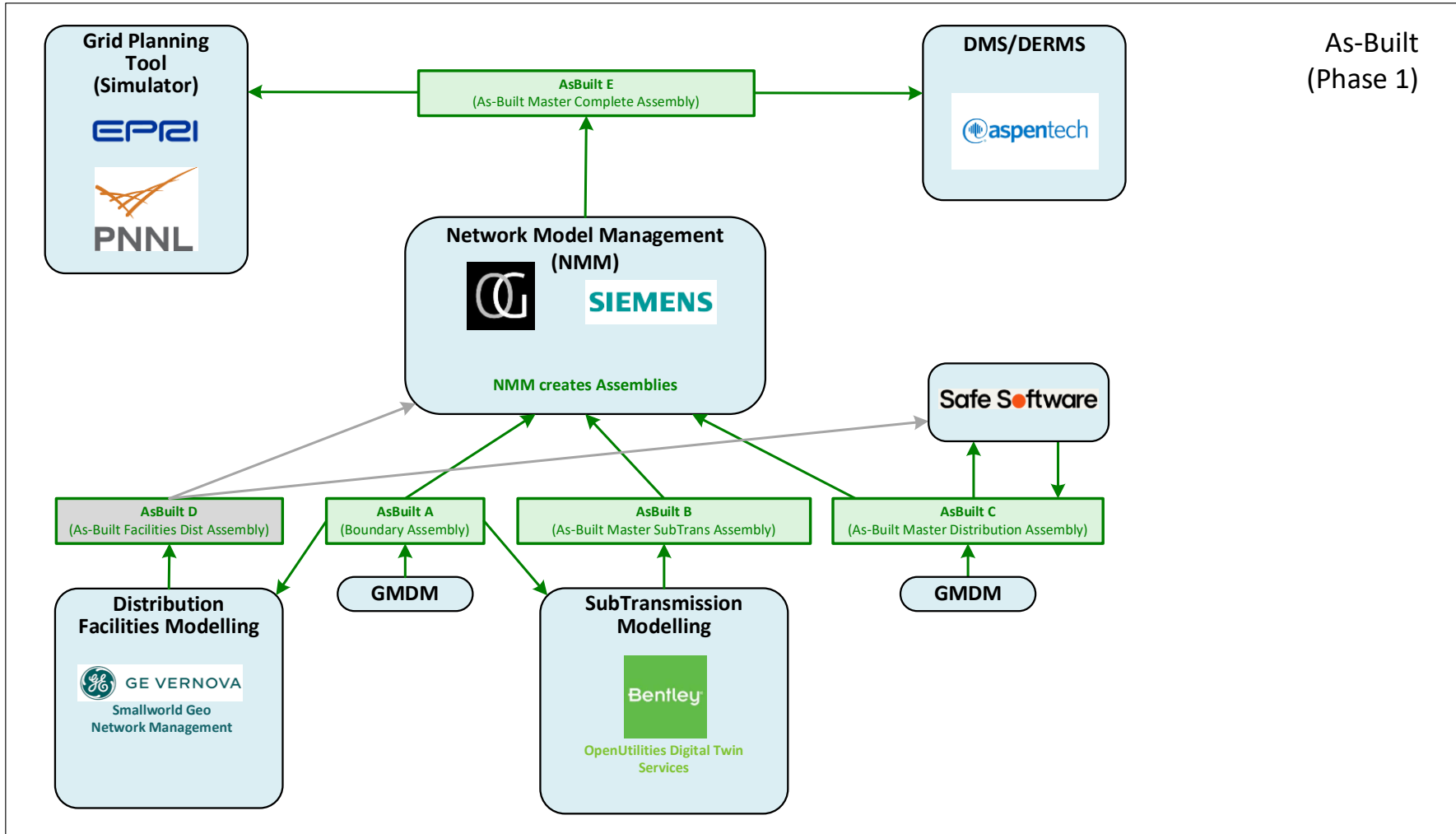
Boundary modelling

The CN Boundary Model (between the SubTransmission portion of the grid and the Distribution portions of the grid) has the following objects.

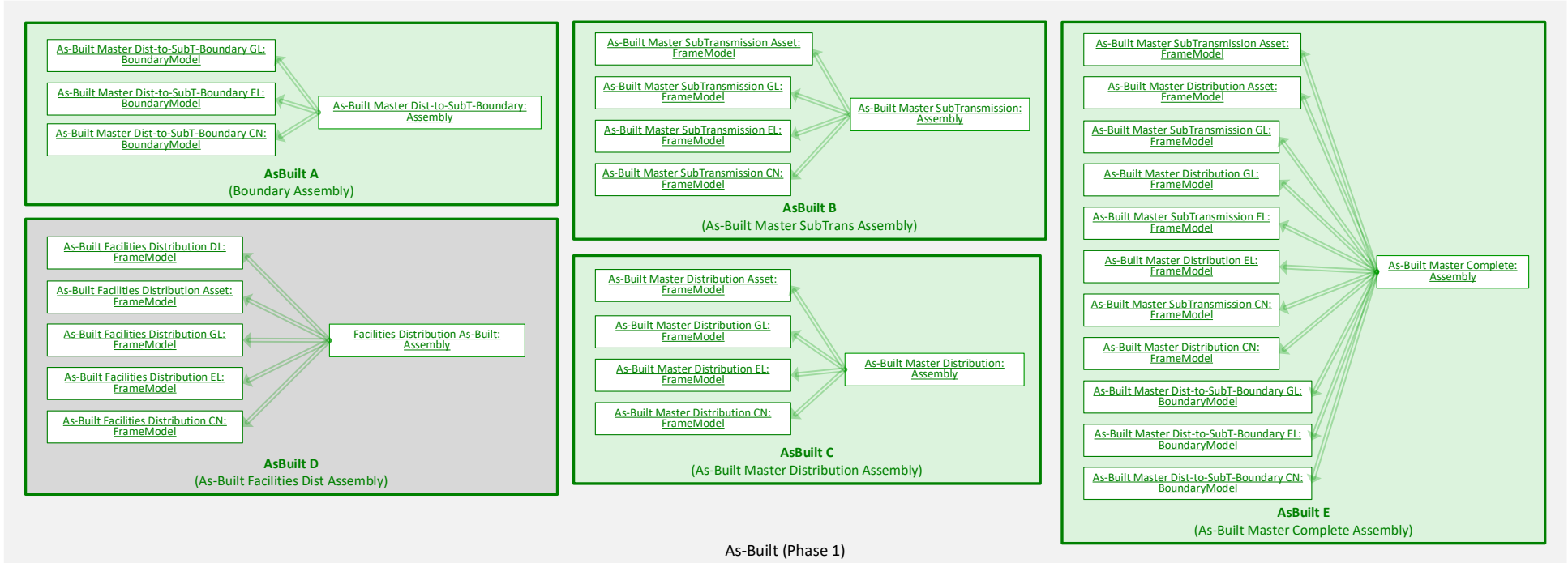
IOP Exchanges

There are two phases to the IOP: As-Built (Phase 1) and Connection Request (Phase 2)

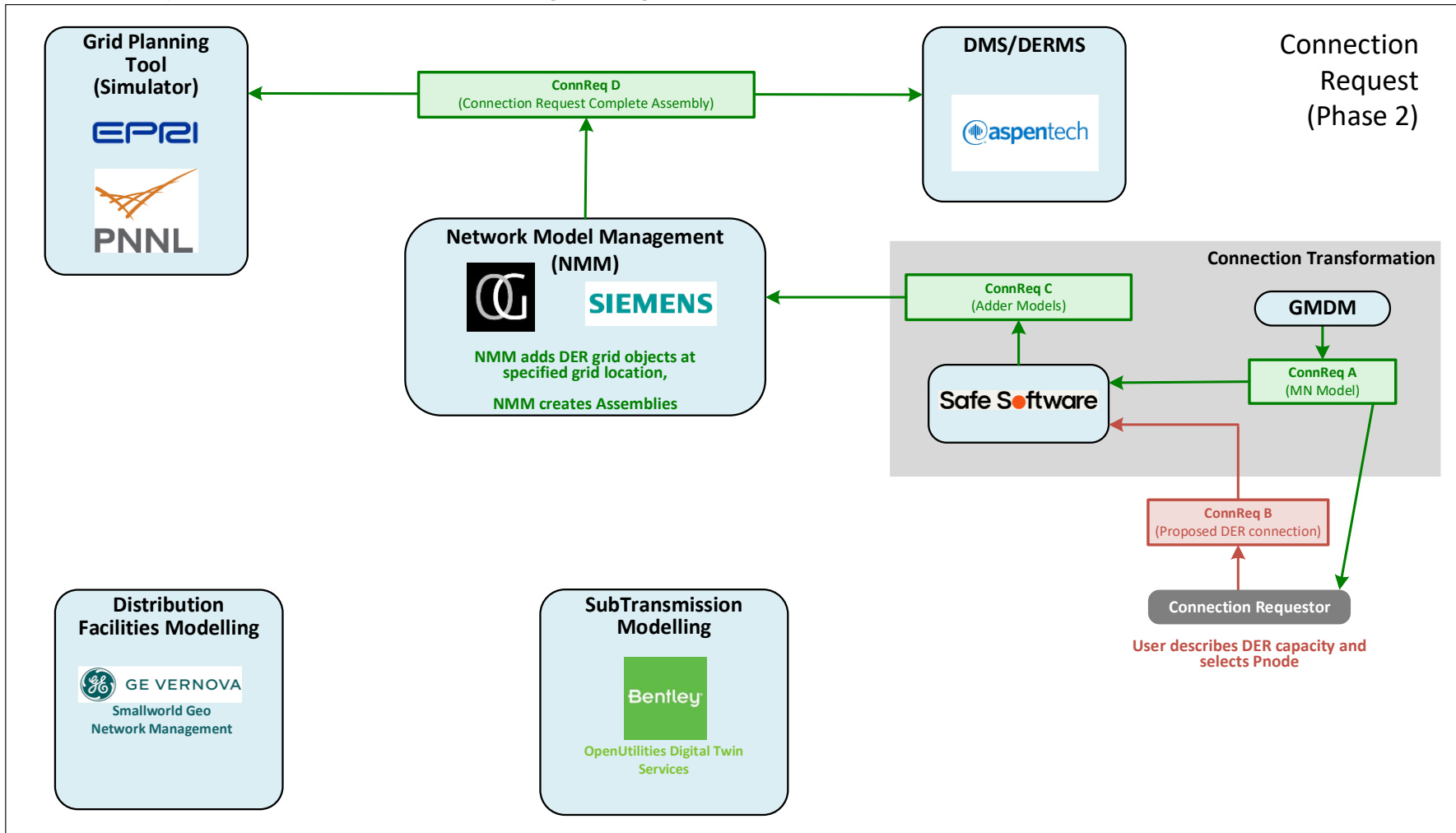
As-Built Phase 1 consists of the following exchanges:



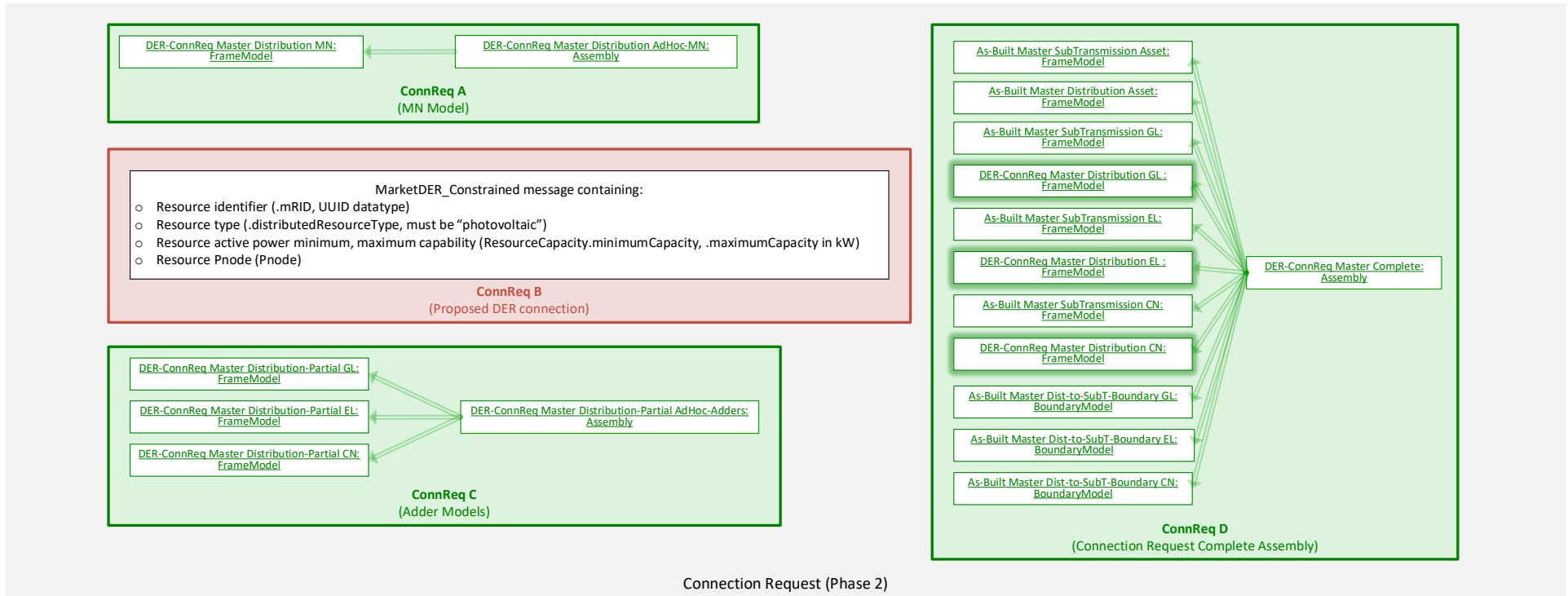
The details of each of the above exchanges is shown below:



Connection Request Phase 2 consists of the following exchanges:



The details of each of the above exchanges is shown below:



Exchange serialization

Models exchanged during the IOP conform to a single profile.

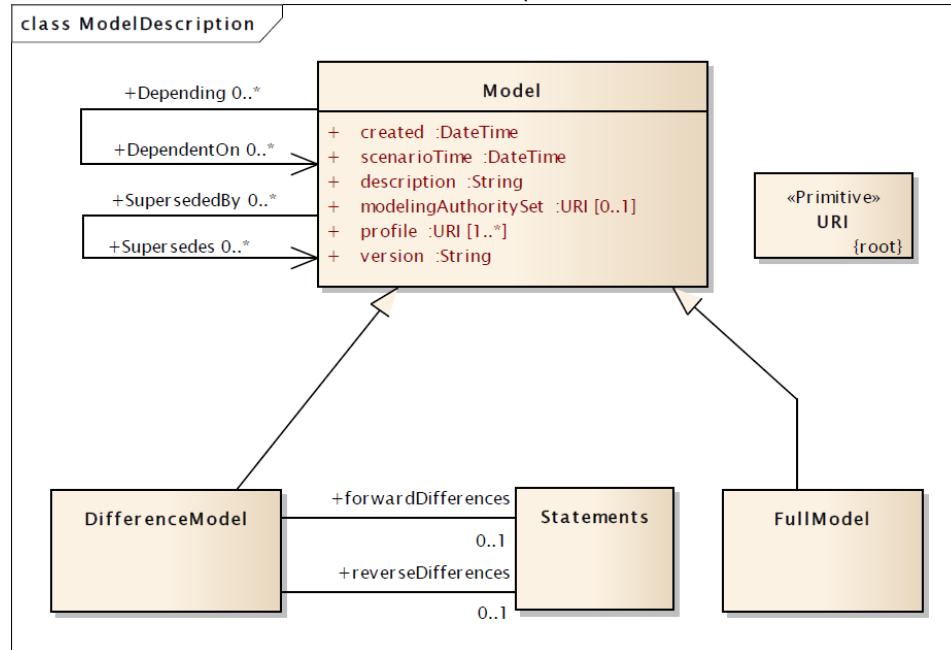
The following profiles are defined for the IOP:

- Connectivity
- Electrical
- GeographicalLocation
- Asset
- DiagramLayout

Model serialization

Serialization generally follows 61970-552 Edition 2, with one profile-conformant Model contained in a .xml file. Metadata about a Model is provided via a FullModel object and via the name of the .xml file.

The information model for the FullModel (as described in 61970-552 Edition 2) is shown below.



GMDM Models utilize all the attributes of FullModel which are inherited from Model, with the exception of

All Model files have a set of declarations at the beginning which identify the xml version and the namespaces referenced in the file. All Models can use the following set of declarations:

```

<?xml version="1.0" encoding="utf-8"?>
<rdf:RDF xmlns:cim="http://cim.ucaiug.io/CIM101/draft#"
  xmlns:rdf="http://www.w3.org/1999/02/22-rdf-syntax-ns#"
  xmlns:gmdm="http://epri.com/gmdm/2025#"
  xmlns:md="http://iec.ch/TC57/61970-552/ModelDescription/1#">

```

A FullModel header object appears as the first resource following the declarations in the file. The FullModel object contains the following properties with the described values:

```

<md:FullModel rdf:about="urn:uuid:UUID"> <!-- unique UUID of this Model: a new UUID is used for each Model instance that is published -->
  <md:Model.created></md:Model.created> <!-- date/time of Model creation in US Eastern time zone in format yyyy-mm-ddThh:mm:ss.nnn-05:00 -->
  <md:Model.scenarioTime>2025-11-04T00:00:00.000-05:00</md:Model.scenarioTime> <!-- date/time represented by Model: fixed -->
  <md:Model.description></md:Model.description> <!-- type of model: see Model metadata table -->
  <md:Model.modelingAuthoritySet></md:Model.modelingAuthoritySet> <!-- portion of grid represented by Model: see Model metadata table -->
  <md:Model.version></md:Model.version> <!-- 001 on first publication of Model for test step, incremented by 1 for each subsequent publication -->
  <md:Model.profile></md:Model.profile> <!-- URI of PSProfileEdition: see Model metadata table -->
</md:FullModel>

```

Additional metadata (test phase & step and Model creator) is provided as part of the .xml file name, whose format is:

<test phase & step>_<Model creator>_<grid object 'family'>_<portion of grid represented>_<profile edition>_<Model version>.xml

Duplicated information (grid object 'family', portion of the grid represented, and profile edition) must match between the .xml name and the FullModel attributes.

Model metadata table

	test phase & step (see Test Procedures for numbering)	Model creator	grid object 'family' used to represent grid	portion of grid represented		profile edition	
In Metadata model	PowerSystemDataSet. businessProcess	GridResour ce. publisher	Model. ModelType	Model.Frame -or- Model.Boundary		Model.PSPProfileEdition	
Using existing - 552 header			md:Model. description		md:Model. modelingAuthoritySet		md:Model.profile
Model .xml file name	1 st field of Model .xml file name	2 nd field of Model .xml file name {Bentley, GE, GMDM, EPRI, OGS, OSI, PNNL, Safe, Siemens}	3 rd field of Model .xml file name	4 th field of Model .xml file name		5 th field of Model .xml file name	
	As-Built.A.x		Master	D-to-SubT-Boundary	http://epri.com/GMDM/D-to-SubT-Boundary	CN	http://epri.com/GMDM_Connectivity/1.0
	As-Built.A.x		Master	D-to-SubT-Boundary	http://epri.com/GMDM/D-to-SubT-Boundary	EL	http://epri.com/GMDM_Electrical/1.0
	As-Built.A.x		Master	D-to-SubT-Boundary	http://epri.com/GMDM/D-to-SubT-Boundary	GL	http://epri.com/GMDM_GeographicalLocation/1.0
	As-Built.B.x		Master	SubTransmission	http://epri.com/GMDM/SubTransmission	CN	http://epri.com/GMDM_Connectivity/1.0
	As-Built.B.x		Master	SubTransmission	http://epri.com/GMDM/SubTransmission	EL	http://epri.com/GMDM_Electrical/1.0
	As-Built.B.x		Master	SubTransmission	http://epri.com/GMDM/SubTransmission	GL	http://epri.com/GMDM_GeographicalLocation/1.0
	As-Built.B.x		Master	SubTransmission	http://epri.com/GMDM/SubTransmission	Asset	http://epri.com/GMDM_Asset/1.0
	As-Built.C.x		Master	Distribution	http://epri.com/GMDM/Distribution	CN	http://epri.com/GMDM_Connectivity/1.0
	As-Built.C.x		Master	Distribution	http://epri.com/GMDM/Distribution	EL	http://epri.com/GMDM_Electrical/1.0
	As-Built.C.x		Master	Distribution	http://epri.com/GMDM/Distribution	GL	http://epri.com/GMDM_GeographicalLocation/1.0
	As-Built.C.x		Master	Distribution	http://epri.com/GMDM/Distribution	Asset	http://epri.com/GMDM_Asset/1.0
	As-Built.D.x		Facilities	Distribution	http://epri.com/GMDM/Distribution	CN	http://epri.com/GMDM_Connectivity/1.0
	As-Built.D.x		Facilities	Distribution	http://epri.com/GMDM/Distribution	EL	http://epri.com/GMDM_Electrical/1.0
	As-Built.D.x		Facilities	Distribution	http://epri.com/GMDM/Distribution	GL	http://epri.com/GMDM_GeographicalLocation/1.0
	As-Built.D.x		Facilities	Distribution	http://epri.com/GMDM/Distribution	Asset	http://epri.com/GMDM_Asset/1.0
	As-Built.D.x		Facilities	Distribution	http://epri.com/GMDM/Distribution	DL	http://epri.com/GMDM_DiagramLayout/1.0
	As-Built.Z.x		Facilities	Complete	http://epri.com/GMDM/Complete	CN	http://epri.com/GMDM_Connectivity/1.0
	As-Built.Z.x		Facilities	Complete	http://epri.com/GMDM/Complete	EL	http://epri.com/GMDM_Electrical/1.0
	As-Built.Z.x		Facilities	Complete	http://epri.com/GMDM/Complete	GL	http://epri.com/GMDM_GeographicalLocation/1.0
	As-Built.Z.x		Facilities	Complete	http://epri.com/GMDM/Complete	Asset	http://epri.com/GMDM_Asset/1.0
	As-Built.Z.x		Facilities	Complete	http://epri.com/GMDM/Complete	DL	http://epri.com/GMDM_DiagramLayout/1.0
	DER-ConnReq.A.x		Master	Distribution	http://epri.com/GMDM/Distribution	MN	http://epri.com/GMDM_MarketNode/1.0
	DER-ConnReq.C.x		Master	Distribution-Adder	http://epri.com/GMDM/Distribution-Adder	CN	http://epri.com/GMDM_Connectivity/1.0

	DER-ConnReq.C.x		Master	Distribution-Adder	http://epri.com/GMDM/Distribution-Adder	EL	http://epri.com/GMDM_Electrical/1.0
	DER-ConnReq.C.x		Master	Distribution-Adder	http://epri.com/GMDM/Distribution-Adder	GL	http://epri.com/GMDM_GeographicalLocation/1.0
	DER-ConnReq.D.x		Master	Distribution	http://epri.com/GMDM/Distribution	CN	http://epri.com/GMDM_Connectivity/1.0
	DER-ConnReq.D.x		Master	Distribution	http://epri.com/GMDM/Distribution	EL	http://epri.com/GMDM_Electrical/1.0
	DER-ConnReq.D.x		Master	Distribution	http://epri.com/GMDM/Distribution	GL	http://epri.com/GMDM_GeographicalLocation/1.0

Examples for a Master SubTransmission Model conforming to version 1.0 of the Electrical profile created by Bentley on the morning of October 31, 2025 for As-Built test step B.1:

- .xml file name :

As-Built.B.1_Bentley_Master_SubTransmission_EL_001.xml

- Declaration and Full Model:

```
<?xml version="1.0" encoding="utf-8"?>
<rdf:RDF xmlns:cim="http://cim.ucaiug.io/CIM101/draft#"
  xmlns:rdf="http://www.w3.org/1999/02/22-rdf-syntax-ns#"
  xmlns:gmdm="http://epri.com/gmdm/2025#"
  xmlns:md="http://iec.ch/TC57/61970-552/ModelDescription/1#">

  <md:FullModel rdf:about="urn:uuid:1c1b1756-4bf7-40dd-8987-98813ea4481a">
    <md:Model.created>2025-10-31T09:27:15.000-05:00</md:Model.created>
    <md:Model.scenarioTime>2025-11-04T00:00:00.000-05:00</md:Model.scenarioTime>
    <md:Model.description>Master</md:Model.description>
    <md:Model.modelingAuthoritySet>http://epri.com/GMDM/SubTransmission</md:Model.modelingAuthoritySet>
    <md:Model.version>001</md:Model.version>
    <md:Model.profile>http://epri.com/GMDM_Electrical/1.0</md:Model.profile>
  </md:FullModel>
```

Assembly serialization

IOP exchanges consist of one or more Models exchanged together. The set of Models in an exchange is classified as an Assembly (if the Models are complete descriptions which are intended to fully represent a grid extent) or an AdHoc collection of Models (if the Models are not). Assemblies and AdHoc collections of Models are shared as .zip files containing the appropriate individual Model .xml files. Metadata for Assemblies and AdHoc collections of Models is provided as part of the .zip file name, whose format is:

```
<test phase & step>_<creator>_<grid object 'family'>_<portion of grid represented>_<date represented>_<type of Model
collection>_<version>.xml
```

NOTE:

- See Assembly and AdHoc collection metadata table below for allowed values for *test phase & step*, *creator*, *grid object family*, *portion of grid represented* and *type of Model collection*
- *date represented* is 2025-11-04
- *version* is 001 on first publication of Assembly for test step, incremented by 1 for each subsequent publication

Assembly and AdHoc collection metadata table

	test phase & step (see Test Procedures for numbering)	Assembly/AdHoc collection creator	grid object 'family' or 'families' used to represent grid	portion of grid represented	Type of Model collection {Assembly, AdHoc-MN, AdHoc-Adders}
In Metadata model	PowerSystemDataSet. businessProcess	GridResource. publisher	Model. ModelType of Models in Assembly	Model.Frame -or- Model.Boundary of Models in Assembly	AssemblyPattern
Assembly .zip file name	1 st field of Assembly .zip file name	2 nd field of Assembly .zip file name {Bentley, GE,GMDM, EPRI, OGS, OSI, PNNL, Safe, Siemens}	3 rd field of Assembly.zip file name	4 th field of Assembly .zip file name	8 th field of Assembly .zip file name
	As-Built.A.x		Master	D-to-SubT-Boundary	Assembly
	As-Built.B.x		Master	SubTransmission	Assembly
	As-Built.C.x		Master	Distribution	Assembly
	As-Built.E.x		Master	Complete	Assembly
	As-Built.D.x		Facilities	Distribution	Assembly
	As-Built.Z.x		Facilities	Complete	Assembly
	DER-ConnReq.A.x		Master	Distribution	AdHoc
	DER-ConnReq.C.x		Master	Distribution-Adder	AdHoc
	DER-ConnReq.D.x		Master	All	Assembly

Example .zip file name for a Master SubTransmission Assembly created by Bentley on the morning of October 31, 2025 for As-Built test step .1:

- .xml file name :
As-Built.B.1_Bentley_Master_SubTransmission_2025-11-04_Assembly_001.xml

(This assembly .zip file would include the example Model .xml file used in the example above.)